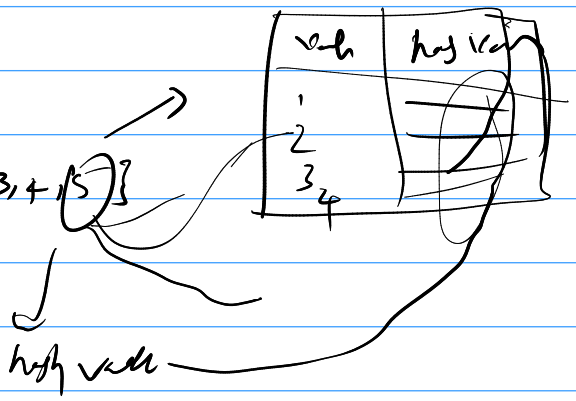
Methods:-

1. add() method:- if you want to add a single element we can you the add

Syntax:-

set.add(elements)

a = {1, 2, 3, 4}
a.add(5)
print(a) {1, 2, 3, 4, 5}



→ update methods

→ if you want to add multiple elements (iterable) mean update()

Syntax

set.update(iterable)

a = {1, 2, 3}
a.update([1, 2, 3, 4, 5, 6])
print(a)

{1, 2, 3, 4, 5, 6}



Remove()

→ Remove is used delete the element on set
→ If the element that you have given not present then it give key error

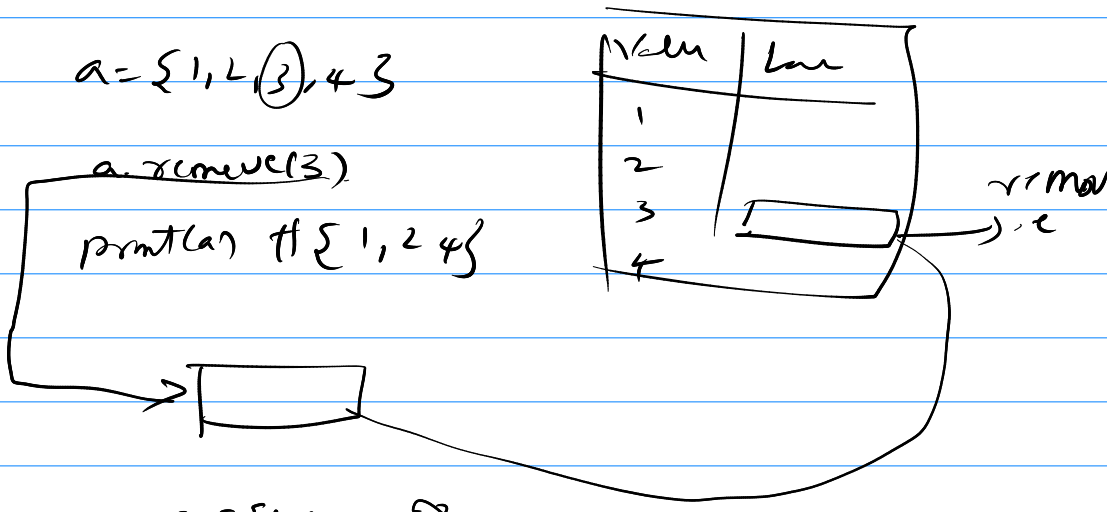
Syntax

set.Remove(element)

a={1,2,3,4}

a.remove(3)

print(a) # {1,2,4}



a={1,2,3,4}
a.remove(5)
print(a) # key error

Discard():-

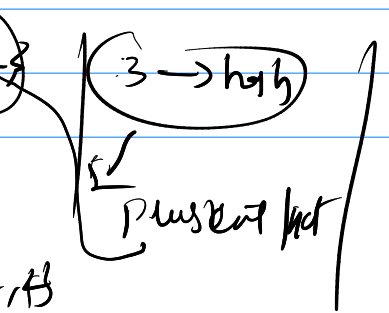
→ It also used to remove the element on set.
→

Syntax

set.discard(element)

→ element ⟶ not present → set present as it is

a={1,2,3,4}
a.discard(3)
print(a)
{1,2,4}



a={1,2,3,4}
a.discard(5) → hash
print(a) {1,2,3,4}

pop() :-

→ it removes random element in a set

syntax

set.pop() → attribute error

s = {1, 2, 3, 4, 5}

s.pop() # Let say 5 is random value

print(s) # {1, 2, 3, 4}

clear() (clear) :-

→ This method used to delete the all elements in set

a = {1, 2, 3, 4}

a.clear()

print(a) # set()

del (keyword)

a = {1, 2, 3, 4}

del a → removed memory obj

print(a) # a not defined.

→ membership

a = [1, 2, 3, 4]

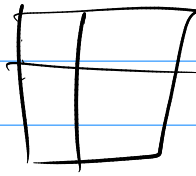
print(4 in a)

True

fast membership

s = {1, 2, 3, 4}

print(4 in s)



True

Set mathematical operations:

1) union → A∪B → | → unique

a = {1, 2, 3, 4}
b = {3, 4, 5, 6}

1 2 3 4 5 6

→ method returns a new set with elements of both sets

Syntax

set.union(set1, set2)

a = {1, 2, 3, 4}
b = {3, 4, 5, 6}
c = a.union(b)
print(c)

{1, 2, 3, 4, 5, 6}

|| → pipe

a = {1, 2, 3, 4}
b = {3, 4, 5, 6}
c = a | b
print(c)

iterable

a = ({1, 2, 3, 4})
b = {3, 4, 5, 6}
c = a.union(b)
print(c)

c = a || b

Intersection(): (A∩B)

→ it returns a new set with similar values in sets

Syntax

set.intersection(set1, set2)

a = {1, 2, 3, 4}
b = {1, 2, 5, 6}

a & b

c = a.intersection(b) => {1, 2}

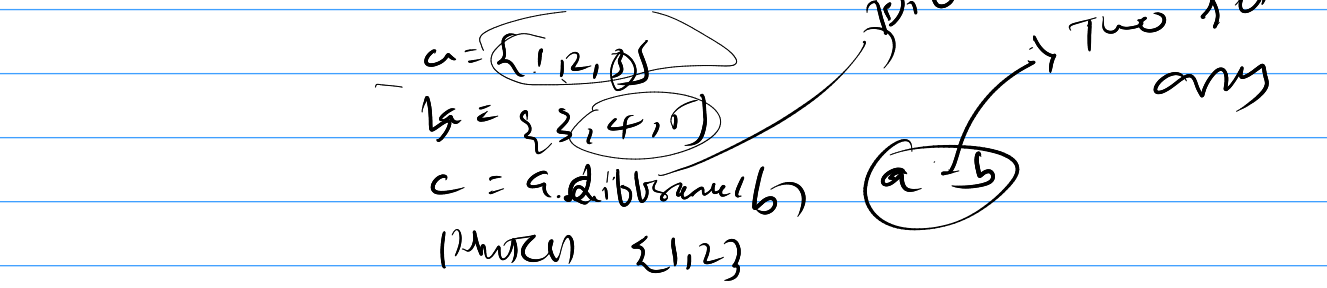
Difference() :-

→ it returns Difference between two sets the returned set contains items contained in first set

-(-)

Syntax

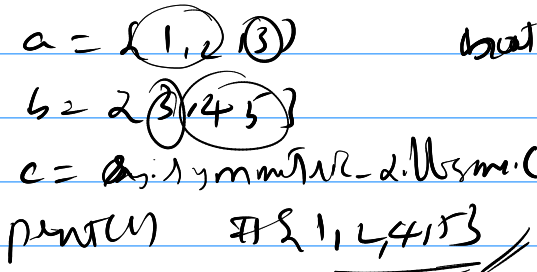
Set.Difference(Set1, Set2...)



Symetric difference:

→ returns a set contains all item from both sets but items not present in set

→



Set Comprehension:-

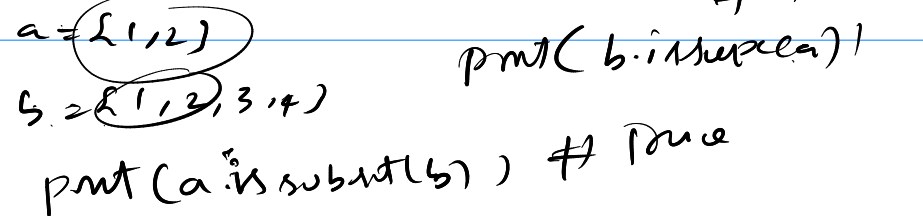
{expression for x in iterable}

a = {i for i in range(1,10)}

print(a)

a = {1, ... 9}

subset



superset {
a → all elements in B
Larger than a

a = {1, 2}
b = {1, 2, 3, 4}
print(b.issuperset(a))
True

* Disjoint set

→ nothing common in b/w sets

a = {1, 2, 7}
b = {8, 5, 6}
print(a.isdisjoint(b)) True

→ ~~intersection~~ intersection_update, difference_update, update, symmetric_difference_update

a = {1, 2, 3, 4} →
b = {3, 4, 5, 6}
a.intersection_update(b)
print(a)

→ it won't return the new set. update existing set.

frozenset:
→ immutable version of set
→ unchangeable, unique, unordered
→ cannot add and remove → can't able to do this

a = frozenset({1, 2, 3, 4})
print(type(a)) <class 'frozenset'>